Interactive 3D Vending Machine

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1. What is the Project About

This project presents an interactive 3D vending machine placed inside a room, developed using OpenGL. The scene contains a vending machine with multiple products, a room environment, and various interactive components. Users can select products using buttons, observe animated dispensing mechanisms, and collect products through an opening collection door. The project demonstrates 3D transformations, hierarchical motion, animation, camera control, and dynamic lighting. Three types of lighting will be used to create realistic illumination and visual effects.

2. Objects Used in the Project

- **Vending Machine:** Main structure containing product shelves, display panels, and control buttons.
- **Products:** Different cans, bottles, and packages placed inside the machine.
- **Product Spirals:** Rotating spiral mechanisms used to move products toward the dispensing area.
- **Dispensing Mechanism:** Components that move to release the selected product.
- **Collection Door:** An opening door through which the dispensed product can be collected.
- **Room Environment:** Floor, walls, ceiling, and other simple objects for the surrounding scene.
- **Interactive Buttons:** Selection buttons used to choose products and control the machine.
- **Indicator Lights:** Dynamic lights showing machine status and product selection.
- **Light Sources:** Three types of lighting to illuminate the room and vending machine.

3. Features and Motions

The project will demonstrate translation, rotation, and scaling of 3D objects. Product spirals will rotate to move products forward, while selected products will move toward the dispensing area. The collection door will open and close using rotation, and related components will move together through hierarchical transformation. Interactive buttons will allow users to select products and operate the vending machine. Indicator lights will change dynamically according to machine activity. Camera controls will provide different views of the scene. The project will also demonstrate three types of lighting, depth testing, perspective projection, and animated objects to create an interactive 3D environment.